Dr B R Ambedkar National Institute of Technology Jalandhar Punjab

Course Title: Computer Graphics and Image Processing Laboratory

Course Code: CSDC0332

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Window-to-Viewport Transformation on an Image

Objective

- To select a portion of an image as a **window** and map it onto a specified **viewport** in order to study image scaling and placement effects in computer graphics.

Learning Outcomes

After performing this experiment, students will be able to:

- understand the concept of **window** and **viewport**
- apply **window-to-viewport transformation**
- observe the effect of **same-size mapping**
- observe **upscaling** when viewport is larger
- observe **downscaling** when viewport is smaller
- analyze image stretching and compression during transformation

Inputs to be Taken

- input image
- window coordinates:
  - x_{wmin}, y_{wmin}
  - x_{wmax}, y_{wmax}
- viewport coordinates:
  - x_{vmin}, y_{vmin}
  - x_{vmax}, y_{vmax}

Algorithm

- Start
- Load the input image.
- Display the original image.
- Take the window coordinates from the user.
- Crop the selected window region from the image.
- Take the viewport coordinates from the user.
- Compute the width and height of the window and viewport.
- Compute scaling factors S_x and S_y .
- Resize the cropped window according to viewport size.
- Create an output canvas.
- Place the transformed image inside the viewport area.
- Display:
  - original image
  - selected window
  - output image with viewport mapping
- Stop.

Tasks to Perform

Task 1: Basic Window-to-Viewport Mapping

Take any image, select a rectangular window, and map it to a viewport at another location.

Task 2: Same Window Size and Same Viewport Size

Choose window and viewport dimensions equal.

Example:

- Window: $(100, 100)$ to $(300, 300)$
- Viewport: $(400, 100)$ to $(600, 300)$

Observation:
Only position changes. Size remains the same. No scaling occurs.

Task 3: Viewport Larger than Window

Choose a small window and a large viewport.

Example:

- Window: $(100, 100)$ to $(200, 200)$
- Viewport: $(350, 50)$ to $(650, 350)$

Observation:
The selected image portion enlarges. This is **upscaling**.

Task 4: Viewport Smaller than Window

Choose a large window and a small viewport.

Example:

- Window: $(50, 50)$ to $(350, 350)$
- Viewport: $(400, 100)$ to $(550, 250)$

Observation:
The selected image portion reduces in size. This is **downscaling**.

Task 5: Different Aspect Ratio Between Window and Viewport

Choose a window and viewport with different width-height ratios.

Example:

- Window size: 200 × 100
- Viewport size: 300 × 300

Observation:
The image gets stretched or compressed because the aspect ratio changes.

Task 6: Compare Three Cases

Perform the experiment for all three cases:

- window size = viewport size
- viewport larger than window
- viewport smaller than window

For each case, calculate and record:

- S_x
- S_y
- type of transformation